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Open Science Arduino OS4P

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1 Introduction

During the coming year, most of us will spend a lot of time at the university doing our thesis. But how good is the air quality actually in different study rooms?

In this experiment, you will find out how to measure the air quality using the Arduino. The idea is to measure the CO₂ concentration, temperature and relative humidity (RH) in each three different rooms and compare the results. According to [1], the CO₂ concentration gives us an indication of how well a room is ventilated. On the other hand, the relative humidity tells us something about how dry or humid the air is [2].

In this document, you will be guided by setting up the Arduino, measuring data with it, processing the data and comparing results to measurements that are done before. In the different sections all of this will be explained in detail so that this document together with the provided codes on GitHub can be used to reproduce the experiment.

The main goal is to reproduce the experiment and to find an answer to the question: which of the buildings has the most healthiest study room?

2 Set up

Firstly, in this Section it will be explained how to build a CO₂ and relative humidity sensor by using the Arduino. The schematic visualization is given in figure 1.

We will now give a step-by-step instruction of how to build the sensor. First we will add the big components to the breadboard.

1. Place the pins of the T/RH sensor (BME/BMP280) into slots g25-g28 of the breadboard.
2. Place the pins of the CO₂ sensor (MH-Z19C) into slots g14-g20.
3. Place the pins of the Micro SD Card Adapter in g2-g7.

Next we will connect these components to the Arduino using jumper cables.

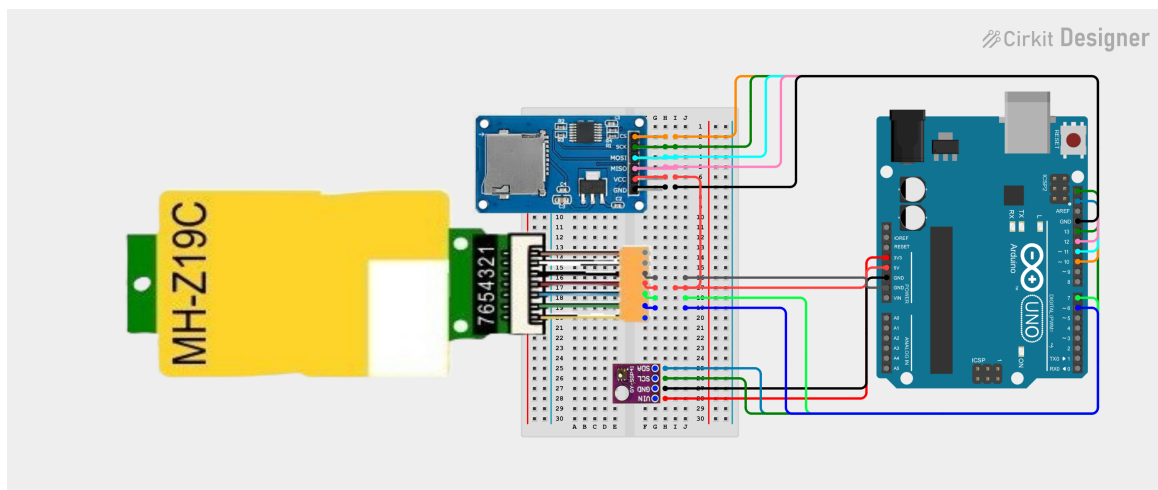


Figure 1: Schematic visualization of the circuit.



Left side of the Arduino (top to bottom)

Arduino	Breadboard
3.3V	h28
5V	j17
GND	h27
GND	j16

Right side of the Arduino (top to bottom)

Arduino	Breadboard
SCL (see bottom)	h26
SDA (see bottom)	h25
GND	i7
13	i3
12	i5
~11	i4
~10	i2
7	j18
~6	j19

3 Experiment

3.1. Test your Arduino

First, you need to download the Arduino IDE program (<https://support.arduino.cc/hc/en-us/articles/360019833020-Download-and-install-Arduino-IDE#installation-instructions>). It is useful to initially test if the arduino works. We can do that by grabbing the usb b to usb c cable and the Arduino. Plug the usb b in the Arduino and the usb c cable in your computer. Open Arduino IDE and select the correct Arduino in the "Select Board" drop down menu. Then open the script under File → Examples → 01.Basics → Blink. Click the rightarrow to upload the code to the Arduino. After a few seconds the onboard LED should start blinking. You can change how fast the LED blinks by changing the number in the delay function in the code and then reuploading it to the Arduino.

3.2. Using the Arduino for measuring CO₂ and relative humidity

Now comes the best and most exciting part, determining which building is the healthiest to study in! Of course, in order to crown the fair winner, all circumstances must be the same. Therefore, we provide you with a plan of how to create the set-up in the study places themselves.

Let's first dive into the environment in which the measurements are done. The locations are as follows:

1. In the KBG building, enter the building through the main entrance. Go to the right and walk past the toilets. You stumble upon a corner with long, white tables where students can study. This is our first location.
2. In the Minnaert building, go to the main hall. There you find the study corners at the windows and tables on the side. Either study cubicle is fine, however keep in mind that the window should be closed if you sit there. Otherwise, the air current could interfere with the measurements, meaning that your results could deviate from ours.
3. In the BBG building, upon entering the building from KBG, immediately go to the left. Here you instantly stumble upon the study space. The Arduino can be set up at either place, as long as the measurements are done in the same space.

Each measurement is done in the second half of the afternoon and took 20 minutes per location, with the Arduino collecting data every 10 seconds. The first measurement was done at 15:30, the second one at 16:00 and the third one at 16:30.



Now for the actual measuring with the Arduino. Connect your Arduino with the right wire setup to your computer and open Arduino IDE three separate times. Make sure the right Arduino board is selected in each window. Put each of the three Arduino codes (Measurements.ino, Read_the_data.ino and Delete.ino) from the Github in a separate window. In order to do 1 measurement you need to:

- First you need make sure the data can be written to the SD card. Upload the Delete.ino code to the Arduino by pressing the right arrow button. It may say the data.csv file is empty already, however we found that we still needed to run this script to collect the data properly.
- Go to your other window and upload the Measurements.ino code to the Arduino. The base time for the experiment in the code is 20 minutes(1,200,000 ms).
- To check if your experiment is running, open Tools → Serial Plotter. After about 20 seconds you should see a graph forming, which updates every 10 seconds. Now simply wait out the 20 minutes being careful not to disturb the measuring devices.
- The data you've collected is stored on the SD card. To collect it go to your window with the Read_the_data code and first open Tools → Serial Monitor. Now upload the Read_the_data.ino code to the Arduino, the data should appear in the Serial Monitor. Click the copy button in the Serial Monitor and paste it into a notepad. Remove possible lines at the top which aren't data and save as a .csv file.
- With your raw data file collected, you can now plot the results. Open for example Spyder and paste the Plot.py code into there. Be sure to save it in the same folder as your raw data file and check if your data variable in the Python will grab the data from the right file. You can now run the code to plot the results and then extract the plots as you wish.

4 Results

The results obtained by following the experimental procedure presented in section 2 and section 3 can be found in the Github. The raw data can be found in the "Data_BBG.csv", "Data_KBG.csv" and "Data_MIN.csv" files. The plots of the data are shown in the "Plot_BBG_data.png", "Plot_KBG_data.png" and "Plot_MIN_data.png" files.

As you might guess, we have accidentally breathed out too close to the sensors a few times leading to a sharp peak in ppm CO₂ and noticable peaks in the temperature and humidity. But looking at the baseline the can conclude a few results. For CO₂ we can see the following: the BBG studyspace has a baseline of 700 ppm CO₂, the Minnaert hall has a baseline of 550 ppm and the KBG study space has a baseline of 500 ppm. The temperature in BBG was around 23.5 degrees Celsius, in Minnaert it was around 24.1 degrees and in KBG the temperature decreased from 23.5 to 22.3 degrees, we are not exactly sure what caused this. The humidity in BBG decreased from 56% to 54%, in Minnaert it decreased from 55% to 52% and in KBG it increased from 54% to 56.5%.

5 Conclusion and outlook

This protocol managed to successfully set up a CO₂, temperature and Relative humidity Arduino-based measuring device. By following the method describe in section 3 we have obtained the plots in section 4. The plots show that the best place to study is KBG (as far as air quality is concerned). It is important to note that our results were measured on one specific day whereas one would desire more data taken over a whole year and many different, non-consecutive days to identify trends in air quality and the overall best place to study. Due to time contains, this experiment is left to the reader. We are confident that with the detailed explanation of the experimental procedure, anyone with a physics background should be able to reproduce and obtain accurate and valuable data to examine the air quality of study spaces at Utrecht University.

Lastly, it is possible that your data might differ from ours. While we do not expect any future measurements to differ significantly from what was presented in this protocol, it is possible that noise, environmental changes and human intervention (ie. cleaning, AC or heating) could alter the air quality at the testing locations in the future.

References

- [1] RIVM. Title of the Web Page. <https://www.rivm.nl/ggd-richtlijn-mmk-ventilatie-scholen>, 2026. Accessed: 2026-09-23. 1
- [2] Anthony V. Arundel, Elia M. Sterling, Judith H. Biggin, and Theodor D. Sterling. Indirect health effects of relative humidity in indoor environments. *Environmental Health Perspectives*, 65:351–361, 1986. URL: <http://www.jstor.org/stable/3430203>. 1